

S&DS 365 / 565
Intermediate Machine Learning

Mercer Kernels Continued

September 21



Reminders

- Assignment 1 out; due October 1 (week from this Thu)
- Quiz 2 end of class Wednesday, material up to (and including) today
- Check Canvas for office hours

Today: Kernels and Neural nets

- ① Recap/discussion of RKHS concepts
- ② Examples (on board)
- ③ Next time(s): basics of neural nets, NTK and double descent

Mercer kernel recap

Summary from last time

- Smoothing methods compute local averages, weighting points by a kernel. The details of the kernel don't matter much
- Mercer kernels using penalization rather than smoothing
- Defining property: Matrix $\mathbb{K}$ is always positive semidefinite
- Equivalent to a type of ridge regression in function space
- The curse of dimensionality limits use of both approaches

Mercer Kernels: The big picture

Instead of using local smoothing, we can optimize the fit to the data subject to regularization (penalization). Choose $\hat{m}$ to minimize

$$\sum_i (Y_i - \hat{m}(X_i))^2 + \lambda \text{penalty}(\hat{m})$$

where $\text{penalty}(\hat{m})$ is a *roughness penalty*.

λ is a parameter that controls the bias-variance tradeoff.

How do we construct a penalty that measures roughness? One approach is: *Mercer Kernels* and *RKHS = Reproducing Kernel Hilbert Spaces*.

What is a Mercer Kernel?

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A Mercer kernel has a special property: For any set of points $x_1, \dots, x_n$ the $n \times n$ matrix

$$\mathbb{K} = [K(x_i, x_j)]$$

is positive semidefinite (no negative eigenvalues)

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This property has many important (and beautiful!) mathematical consequences. It is a characterization of Mercer kernels.

Mercer Kernels: Key example

A Gaussian gives us a Mercer kernel:

$$K(x, x') = \frac{1}{h} e^{-\frac{(x-x')^2}{2h^2}}$$

Note: Here we fix the bandwidth h .

Another example

The triangular kernel:

$$T(u) = (1 - |u|)_+$$

and then we define

$$K(x, x') = \frac{1}{h} T\left(\frac{|x - x'|}{h}\right)$$

Basis functions

We can create a set of *basis functions* based on K .

Fix z and think of $K(z, x)$ as a function of x . That is,

$$K(z, x) = K_z(x)$$

is a function of the second argument, with the first argument fixed.

Defining an inner product (geometry)

Because of the positive semidefinite property, we can create an *inner product* and *norm* over the span of these functions

If $f(x) = \sum_i \alpha_i K_{x_i}(x)$, $g(x) = \sum_i \beta_i K_{x_i}(x)$, the inner product is

$$\begin{aligned}\langle f, g \rangle_K &= \sum_i \sum_j \alpha_i \beta_j K(x_i, x_j) \\ &= \alpha^T \mathbb{K} \beta\end{aligned}$$

where $\mathbb{K} = [K(x_i, x_j)]$

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The norm is

$$\begin{aligned}\|f\|_K^2 &= \langle f, f \rangle_K = \sum_i \sum_j \alpha_i \alpha_j K(x_i, x_j) \\ &= \alpha^T \mathbb{K} \alpha \geq 0\end{aligned}$$

In fact $\|f\|_K = 0$ if and only if $f = 0$ (see notes)

Assignment 1 will solidify your understanding of Mercer kernels and kernel ridge regression!

Reducing to finite dimensions

Representer Theorem

Let $\hat{m}$ minimize

$$J(m) = \sum_{i=1}^n (Y_i - m(X_i))^2 + \lambda \|m\|_K^2.$$

Then

$$\hat{m}(x) = \sum_{i=1}^n \alpha_i K(X_i, x)$$

for some $\alpha_1, \dots, \alpha_n$.

So, we only need to find the coefficients

$$\alpha = (\alpha_1, \dots, \alpha_n).$$

Gradient descent

The gradient descent updates to α are

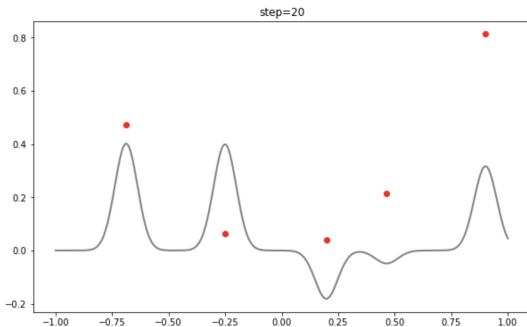
$$\alpha \longleftarrow \alpha + \eta (\mathbb{K}(\mathbf{y} - \mathbb{K}\alpha) - \lambda \mathbb{K}\alpha)$$

where $\mathbb{K}$ is the $n \times n$ Gram matrix and $\eta > 0$ is a step size.

Demo

```
if step % 10 == 0:  
    plot_function_and_data(x, f, X, y, t=step, sleeptime=.5,  
alpha = alpha + stepsize * (K.T @ (y - K @ alpha) - lam * K
```

6] 6.4s



Kernels from features—and vice-versa

If $x \rightarrow \varphi(x) \in \mathbb{R}^d$ is a feature mapping, we can define a Mercer kernel by

$$K(x, x') = \varphi(x)^T \varphi(x')$$

Conversely, for any Mercer kernel we can derive the corresponding feature map (from the spectral theorem)

Kernels are central in ML

- Mercer kernels play a central role in machine learning
 - ▶ Can define similarity functions that are kernels for all kinds of data — graphs, molecules, text documents
 - ▶ Gaussian processes
 - ▶ Modern understanding of deep neural networks

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